Lecture 1: Introduction to Psychology (PSY 101)

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Class Norms and Expectations

1. **Attendance & Participation**
    - Being present in class regularly is important because psychology often involves discussions, reflections, and participation-based learning activities.
2. **Open-mindedness and Respect**
    - Since psychology explores human thoughts, emotions, and behavior, everyone may have diPerent perspectives.

You must respect diPering views and avoid judgment.

3. **Encouragement to Question and Reflect** ○ Psychology thrives on curiosity. Asking *"why"* behind behaviors and mental processes helps deepen understanding.
4. **Confidentiality**
    - During class discussions or personal sharing, maintaining privacy is essential to build trust and respect among peers.

Self-Reflection Before Beginning

Ask yourself:

- Why am I taking this course? (Interest, career, self-understanding?)
- What do I hope to learn? (About the brain, behavior, emotions?)
- How do I learn best? (Visually, auditorily, in groups, or independently?)

Self-awareness of your learning style can help you succeed in this course — just as psychology emphasizes self-understanding in general.

Course Overview

Psychology helps us understand **how and why humans think, feel, and behave** the way they do.

This course introduces both **scientific methods** and **theoretical foundations** of the field.

Major Topics:

1. **Introduction to Psychology** – What it is and why it matters.
2. **The Brain** – Structure and biological functions aPecting behavior.
3. **Perception** – How we interpret sensory information.
4. **Cognition** – Thinking, reasoning, memory, and language.
5. **Emotion** – Feelings and their impact on actions.
6. **Personality** – Individual diPerences and traits.
7. **Development** – Growth from infancy to adulthood.
8. **Psychopathology** – Psychological disorders and mental illness.
9. **Social & Group Interactions** – How we behave with others in groups and society.

Definition of Psychology

"Psychology is the **scientific study of behavior and mental processes.**"

Let's break this down:

Term	Meaning
Scientific	Psychology uses **systematic, objective, and empirical methods** (e.g., experiments, observations, surveys) to gather verifiable data.
Behavior	Refers to **observable actions** like talking, walking, crying, eating — anything we can see or measure.
Mental Process	Refers to internal experiences such as thoughts, emotions, memory, and motives that cannot be directly observed but can be inferred from

Example:

If someone suddenly cries, psychology tries to understand both the **behavior** (crying) and the **mental process** (sadness, loss, or stress) behind it.

History of Psychology

Ancient Origins

Humans have always wondered about the mind and soul.

- In **Ancient Egypt**, the *Edwin Smith Papyrus* (around 1600 BCE) contained one of the earliest recorded observations about the **brain's role** in human functioning.

- Philosophers such as **Plato** and **Aristotle** in ancient Greece also debated the connection between the mind and body.

Birth of Experimental Psychology

- **1854 – Gustav Fechner (Germany):**

Developed the **first theory** on how humans make judgments about sensory experiences.

→ His work laid the foundation for **Signal Detection Theory**, which explains how we detect faint signals amid background noise.

- **1879 – Wilhelm Wundt (Germany):**

Founded the **first psychology laboratory** in Leipzig.

→ Marked the oPicial beginning of psychology as a **scientific discipline.**

→ Wundt focused on **introspection**, where people described their conscious experiences in detail.

→ He was the **first person to identify as a psychologist.**

Development and Global Spread

- **G. Stanley Hall (1880s):**
    - Established the first psychology lab in the **United States**.
    - Promoted **scientific pedagogy** — applying psychology to education.
- **John Dewey (1890s):**
    - Developed **educational theories** that emphasized **learning by doing** and adapting lessons to real-life experiences.
    - Considered one of the fathers of modern education.

Behaviorism (Early 1900s)

- Founded by **John B. Watson**, later expanded by **B. F. Skinner**.
- **Main Focus:** Study only **observable and measurable behavior** — not thoughts or feelings.
- Introduced the idea of **conditioning**:
    - **Classical Conditioning:** Learning through association (Pavlov's dogs).
    - **Operant Conditioning:** Learning through **rewards and punishments** (Skinner).

Example:

A student studies hard to get praise (reward) → increases studying behavior.

Cognitive Revolution (Mid to Late 20th Century)

- Psychologists began to realize that **mental processes** (thinking, memory, problem-solving) were crucial.
- This led to the rise of **Cognitive Science**, combining multiple disciplines:
  - Cognitive Psychology
  - Linguistics
  - Computer Science
  - Philosophy
  - Neurobiology

Focus: Understanding how we process information, think logically, store memories, and make decisions.

Major Perspectives in Psychology

Each perspective provides a diPerent way to explain human behavior:

Perspective	Focus Area	Key Assumption / Example
Biological	Brain, hormones, and genetics	Behavior is influenced by biological structures and chemical processes. *Example:* Depression may be caused by low serotonin levels.
Behavioral	Learning from environment	Behavior is shaped through conditioning and reinforcement. *Example:* A child learns to say "thank you" after being praised.

Perspective	Focus Area	Key Assumption / Example
		Focuses on how people think, understand, and

Cognitive	Mental processes	remember. *Example:* Problem-solving or decisionmaking strategies.
Psychoanalytic	Unconscious motives	Founded by Freud; behavior is driven by unconscious desires and childhood conflicts. *Example:* Phobias may reflect repressed fears.
Humanistic	Free will and growth	Emphasizes human potential, self-actualization, and personal growth. *Example:* People strive to become the best version of themselves.
Socio-cultural	Society and culture	Behavior is influenced by social norms, traditions, and cultural expectations. *Example:* Collectivist vs. individualist cultures affect family roles.
Evolutionary	Adaptation and survival	Human traits evolved to enhance survival and reproduction. *Example:* Fear of snakes may be an evolved protective response.

Perception, Thinking, and the Stroop Efect

Automaticity

- When tasks become so familiar that they require **little conscious efort**.

Example: Reading is automatic — once you learn it, you don't need to think about every letter.*

Stroop Efect

- Demonstrates **interference between automatic and controlled processes.**
- Task: Name the **color** of the ink in which a word is printed rather than reading the word itself.

Example: The word "RED" written in blue ink.*

- People take longer or make mistakes because **reading (automatic)** interferes with **color recognition (controlled)**.
- This experiment shows that **attention** and **processing speed** aPect performance.

Mental Health (According to WHO, 2022)

"Mental health is a state of mental well-being that enables people to cope with life's stresses, realize their abilities, learn well and work well, and contribute to their community."

Key Takeaways:

1. Mental health ≠ just the absence of mental illness.

→ It also means **positive psychological functioning.**

2. Includes **emotional**, **psychological**, and **social well-being**.

→ How we feel, think, and relate to others.

3. Influenced by **individual, social, and structural factors** — such as genetics, environment, and community support.
4. Mental health can **change over time**, depending on life circumstances.

→ For instance, stress, loss, or trauma can aPect well-being.

Objectives of Psychology

Psychology aims to **understand and explain human thought and behavior** through the following:

- **Describe** how people behave.
- **Explain** why they behave that way.

- **Predict** future behavior based on observation.
- **Control or influence** behavior positively (e.g., therapy, education, workplace training).

It studies:

- How our **minds interpret sensory information** (what we see, hear, taste, feel).
- How we **think**, **learn**, and **remember**.
- How we **interact socially** and **make moral or emotional decisions**.

Subfields of Psychology

Subfield	Focus Area	Example
Biological Psychology	Links between brain, nervous system, and behavior	Studying how brain injuries aPect memory.
Clinical Psychology	Diagnosis and treatment of mental disorders	Therapy for anxiety or depression.
Cognitive Psychology	Study of mental processes like perception, memory, reasoning	Research on how memory works.
Developmental Psychology	Growth and change across lifespan	Studying child development or aging.
Educational Psychology	Learning and teaching methods	Designing ePective classroom strategies.
Social Psychology	Behavior in groups and social influence	Studying conformity or prejudice.
Industrial-Organizational Psychology	Behavior in workplace settings	Improving employee motivation and productivity.
Health Psychology	Connection between mind and physical health	Research on stress and its ePects on the immune system.
Counseling Psychology	Helping individuals achieve personal well-being	Providing guidance for relationship or career issues.

In Summary

- **Psychology** = Science of **behavior and mental processes**.
- It began as **philosophy**, became a **science** with Wundt in 1879, and evolved through **behaviorism** and **cognitive science**.
- It includes multiple **perspectives** (biological, behavioral, cognitive, etc.), giving diPerent explanations for human behavior.
- It studies not only **mental illness** but also **positive mental health**, aiming to improve life quality.
- Psychology applies to **education, health, relationships, workplaces, and society** as a whole.

Lecture 02: Research Methods in Psychology

"Asking Questions, Getting Answers"

1. What is Research in Psychology?

Definition

Psychological research is the **systematic, scientific study of behavior and mental processes**.

It uses **empirical methods** (observation, measurement, and experimentation) to understand how people think, feel, and act.

Psychological research aims to **develop theories**, **test hypotheses**, and **apply findings** to real-life problems such as mental health, education, workplace behavior, and social issues.

Objectives of Psychological Research

Goal	Description	Example
Description	To observe and describe behavior as it occurs naturally.	Observing how children interact in a playground.
Explanation	To determine **why** behavior occurs.	Studying whether aggression in children is caused by frustration.
Prediction	To predict how people will behave in certain situations.	Predicting that sleep deprivation leads to lower academic performance.
Control	To influence or modify behavior.	Using therapy to reduce anxiety.

Importance of Research in Psychology

1. **Evidence-based Understanding:** Replaces myths and opinions with tested knowledge.
2. **Improves Human Welfare:** Research findings improve education, health, business, and social policy.
3. **Theory Development:** Theories are built, tested, refined, or rejected through research.
4. **Scientific Progress:** Leads to innovation in therapy, learning methods, and diagnostics.

5. **Informs Practice:** Guides professionals like counselors, teachers, and clinicians.

Example:

Early research on **classical conditioning (Pavlov)** and **operant conditioning (Skinner)** laid the foundation for behavioral therapies still used today.

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2. What is Critical Thinking?

Definition:

Critical thinking is the **intentional process of analyzing and evaluating information objectively** in order to make reasoned and evidence-based judgments.

It means not accepting statements at face value — instead, questioning assumptions, testing evidence, and considering alternative explanations.

Stephen D. Brookfield's (1987) Components of Critical Thinking:

1. **Identifying and Challenging Assumptions** o Recognize hidden biases, values, and beliefs behind arguments.
   - o Example: "Men are naturally better at math" — a critical thinker questions whether this stereotype has scientific support.
2. **Recognizing the Importance of Context** o Behavior and beliefs vary across situations and cultures. o Example: Eye contact may show respect in one culture but rudeness in another.
3. **Imagining and Exploring Alternatives** o Considering multiple explanations or solutions.
   - o Example: A student scoring poorly might be unmotivated — or anxious, tired, or facing personal challenges.
4. **Reflective Skepticism** o Withholding belief until suPicient evidence is available. o Example: Questioning viral claims like "Listening to Mozart makes babies smarter."

Why Critical Thinking is Essential in Psychology

- Prevents blind acceptance of "common sense" or "folk psychology."

- Encourages evidence-based reasoning.
- Helps evaluate research validity and reliability.
- Improves decision-making and ethical reasoning.

- Encourages **scientific skepticism**, not cynicism — open-minded but cautious thinking.

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3. What Makes Something Scientific?

A study is **scientific** when it adheres to the **scientific method**, ensuring reliability, objectivity, and empirical testing.

Core Principles of Scientific Inquiry

1. **Empiricism:**

Knowledge must come from **observation and evidence**, not opinion.

2. **Falsifiability:**

A theory must be **testable and disprovable** (Popper, 1959).

Example: “All swans are white” is falsifiable — one black swan disproves it.

3. **Replication:**

Findings must be **repeatable** by other researchers.

4. **Objectivity:**

Personal bias must be minimized through standardized procedures.

5. **Systematic Observation:**

Data must be gathered in a structured, methodical way.

6. **Quantifiability:**

Concepts should be measurable (using operational definitions).

7. **Probabilistic Reasoning:**

Psychological research deals with probabilities, not certainties — human behavior varies across contexts.

Example: Wundt’s First Psychological Experiment

Wilhelm Wundt (1832–1920), the “father of experimental psychology,” studied **reaction time** — how long it takes a person to respond to a stimulus.

He found that making a decision (red or green light) took longer than a simple reflex action — showing that mental processes could be **measured scientifically**.

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4. Steps in the Research Process

1. **Identify a Problem / Research Question** o Example: “Does social media use cause anxiety in teenagers?”

2. **Review the Literature** o Study previous research to identify gaps and avoid repetition.

3. **Formulate a Hypothesis** ○ A **testable statement** predicting the relationship between variables.

   ○ Example: “Increased social media use leads to higher anxiety levels.”

4. **Select Research Method** o Choose design: experimental, correlational, survey, case study, etc.

5. **Collect Data** o Use valid tools: tests, interviews, surveys, observations.

6. **Analyze Results** o Apply statistical or qualitative analysis to test hypotheses.

7. **Report Findings** o Share results through journals, conferences, or reports.

8. **Replication** o Others repeat the study to verify results.

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5. Types of Research in Psychology

A.

Experimental (Causal) Research

Purpose:

To establish **cause-and-efect** relationships.

Structure:

- **Independent Variable (IV):** Manipulated factor.
- **Dependent Variable (DV):** Measured outcome.
- **Control Group:** No treatment.
- **Experimental Group:** Receives treatment.

Example:

Testing whether caPeine (IV) improves memory (DV).

Control: Participants receive no caPeine.

Strengths:

- Establishes causality.
- High control and internal validity.

Limitations:

- Artificial settings reduce realism.
- Not always ethical (e.g., studying trauma).

B.

Non-Experimental Research

Definition:

No variable manipulation — researcher observes naturally occurring behavior.

Purpose:

Used when experiments are impossible or unethical.

Examples:

- Studying ePects of parental divorce on children.
- Analyzing historical data (e.g., post-war mental health trends).

Strengths:

- Real-world context.
- Can explore phenomena over long periods.

Limitations:

- No control over variables.
- Cannot infer causality.

C.

Descriptive Research

Goal:

To describe characteristics or behaviors in detail.

Methods:

2. **Case Study:** ○ Deep analysis of a single individual or group.

- Example: Freud's patient studies (e.g., "Little Hans").

3. **Naturalistic Observation:**
   - Observing behavior in its natural environment.
   - Example: Watching playground interactions.
4. **Survey Research:**
   - Using questionnaires to collect large-scale data.

Strengths:

- Provides detailed descriptions.

Weakness:

- Cannot explain causes.

D.

Correlational Research

Goal:

To measure the **relationship between variables**.

Example:

Studying the relationship between stress and academic performance.

Types of Correlation:

- **Positive:** Both variables increase (e.g., height and weight).
- **Negative:** One increases, the other decreases (e.g., stress and sleep).
- **Zero:** No relationship.

Important:

Correlation does not imply causation.

Two variables may be related because of a third variable (confounding variable).

E.

Quasi-Experimental Research

Definition:

Similar to experiments but without random assignment.

Example: Comparing two existing school classes taught with diPerent methods.

Used When:

Ethical or practical issues prevent random assignment.

F.

Longitudinal vs Cross-Sectional Studies

Type	**Description**	**Example**	**Advantage**	**Limitation**
Longitudinal	Studies same group over years	Tracking memory decline in aging adults	Shows changes over time	Timeconsuming & expensive
Cross-sectional	Compares diPerent groups at one time	Comparing teens & adults on sleep habits	Quick & inexpensive	No cause-ePect or time analysis

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6. **Problems and Biases in Research**

1. **Sampling Bias:**
   - Participants don't represent population.
   - Example: Only college students used → results not generalizable.
2. **Experimenter Bias:**
   - Researcher's expectations subtly influence outcomes.
3. **Demand Characteristics:**
   - Participants act how they think they should.
4. **Confirmation Bias:**
   - Researcher interprets data to fit their hypothesis.
5. **Placebo Efect:**
   - Participants improve because they believe they received treatment.
6. **Ethical Concerns:**
   - Must protect participants:
     - Informed consent
     - Confidentiality
     - Right to withdraw
     - No physical or psychological harm

7. Quantitative vs Qualitative Research

Feature	Quantitative Research	Qualitative Research
Focus	Numbers and measurements	Meanings and experiences
Goal	Test hypotheses, generalize results	Explore depth, understand context
Approach	Objective, structured	Subjective, open-ended
Data Type	Numerical (scores, scales)	Textual (words, images)
Analysis	Statistical	Thematic / narrative
Example	Survey on stress levels	Interview on coping experiences
Strength	Reliable, generalizable	Deep, detailed insights
Weakness	May ignore context	Hard to generalize

Quantitative Methods Include:

- Experiments
- Surveys
- Questionnaires
- Database analysis
- Statistical modeling

Qualitative Methods Include:

- In-depth interviews
- Focus group discussions
- Participant observations
- Case studies
- Document analysis

8. Folk Psychology vs Scientific Psychology

Folk Psychology:

Refers to everyday beliefs and “common sense” ideas people have about behavior and the mind.

These are often influenced by culture, experience, and intuition rather than science.

Scientific Psychology:

Uses the scientific method to test ideas with evidence and reject unsupported claims.

Common Myths in folk psychology

Claim	Scientific Finding
Opposites attract	Similarity predicts attraction.
Familiarity breeds contempt	Familiarity usually increases liking.
More people = more help	Bystander ePect: people are less likely to act.
Visual vs verbal learners	Learning styles theory lacks evidence.
Hypnosis is fake	Has real psychological ePects in pain reduction and therapy.
Subliminal ads work	No strong evidence.
Mozart makes babies smarter	Only short-term ePects.
Old age = unhappiness	Older adults often report higher life satisfaction.
Trust your first answer	Changing answers can improve accuracy.
Stress causes ulcers	Caused by bacteria (H. pylori).
Positive thinking cures cancer	No direct causal link.
Parenting solely determines personality	Genetics and peers also shape traits.
Inkblot tests reveal personality	Low reliability and validity.
Interviews predict job success	Structured tests are better predictors.

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9. Cultural Reflection: Psychology Beyond the West

The lecture ends by reminding us that most “folk psychology” ideas come from the **Western world**.

Students are encouraged to **reflect on local cultural beliefs** — for example:

- “Left-handed people are unlucky.”
- “Dreams always predict the future.”
- “Mental illness is caused by evil spirits.”

Psychology encourages evaluating such beliefs **scientifically**, not dismissively — combining **critical thinking** with **cultural awareness**.

FINAL SUMMARY TABLE (Exam Quick Revision)

Topic	**Core Idea**	**Key Example**
Critical Thinking	Evaluate evidence & assumptions	Questioning myths about mental illness
Scientific Method	Systematic, testable, falsifiable	Wundt's reaction time experiment
Experimental Research	Manipulate variables to find cause	EPect of caPeine on memory
Non-Experimental	Observe natural situations	Studying stress during exams
Descriptive	Describe behavior	Case study on Phineas Gage
Correlational	Measure relationships	Stress vs GPA
Quantitative	Numerical data, statistical	Survey data analysis
Qualitative	Narrative data, meanings	Interview about anxiety
Folk Psychology	Common myths	"Opposites attract"
Scientific Psychology	Evidence-based	Similarity leads to liking

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LECTURE 03-THE BIOLOGY OF MIND

1. What is the Mind?

Do you have a mind? What is it? Where is it?

- The mind is not an object you can touch.
- The mind is what your brain does.
- It is created through the brain's information processing.

Functions of the Mind

Your mind enables:

- Thinking
- Decision-making
- Memory
- Emotions
- Imagination
- Perception (what you see, hear, feel, taste, smell)

In simple words:

Your mind is your inner world, created by the brain's activity.

2. The Brain – The Central Command Center

The brain is the control system that regulates nearly everything in the body.

Major functions of the brain

- Thoughts (reasoning, planning, problem-solving)
- Memory (storing and recalling information)
- Learning (forming new connections between neurons)
- Emotions (fear, happiness, anger, love)

- Movement (coordination, balance)
- Senses (interpreting visual, auditory, taste, touch, smell signals)
- Automatic functions
- Heartbeat
- Breathing
- Digestion
- Sleep cycles
- Stress response
- Hormone regulation (e.g., puberty)

Mini-diagram (described):

Imagine a control center filled with switches — each switch manages one body function. That control center is your brain.

3. The Nervous System

The nervous system is the body's communication highway.

Two main parts:

1. Central Nervous System (CNS)
   - Brain
   - Spinal cord
2. Peripheral Nervous System (PNS)
   - All nerves outside the brain and spinal cord
   - Connects the CNS to the body (muscles, organs, skin)

Purpose:

To send messages rapidly throughout the body using electrical and chemical signals.

4. Neurons – The Brain's Building Blocks

- A neuron is a nerve cell that sends and receives information.
- You have ~86 billion neurons in your brain.

Parts of a neuron

- Dendrites – receive messages
- Cell body (soma) – processes information
- Axon – sends messages
- Synapse – tiny gap where chemical signals pass to the next neuron

Example

When you touch a hot cup:

- Sensory neurons send a pain signal → brain
- Brain processes it
- Motor neurons send message → move your hand away

5. The Brain's Organization (3 Major Parts)

A. The Brainstem

The oldest and most basic part of the brain.

Controls survival functions.

Parts of the Brainstem

1. Medulla – controls heartbeat and breathing
2. Pons – controls sleep, facial expressions, and movement coordination
3. Reticular formation – controls alertness and arousal

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4. Thalamus – the "sensory relay station"

- Sends sensory information (except smell) to higher brain regions

Mini-diagram (described):

A stalk-like structure at the bottom of the brain — that’s the brainstem.

B. The Limbic System

The emotional center of the brain.

Key Structures

1. Amygdala – emotion (especially fear and aggression)

- Example: Your amygdala activates when you see a snake.

2. Hippocampus – memory formation

- Example: Helps you remember your first day of school.

3. Hypothalamus – regulates hunger, thirst, temperature, hormones

- Commands the pituitary gland (master hormone gland)

4. Pituitary gland – releases hormones controlling growth, stress, reproduction

Mini-diagram (described):

Small structures deep inside the brain forming a ring — that is the limbic system.

C. The Cerebral Cortex

The newest, most advanced part of the brain.

Responsible for complex thinking, planning, language, and consciousness.

Functions

- Logic
- Creativity
- Language
- Moral judgement
- Planning

- Problem-solving

6. Case Study: Phineas Gage

The most famous case in neuroscience.

The Accident

- In 1848, a metal rod shot through Gage's cheek and out the top of his skull.
- It destroyed much of his frontal lobes.

What happened after?

- He survived and could speak, walk, and think.
- BUT his personality changed completely:
- Became irritable
- Impulsive
- Dishonest
- Lost ability to manage emotions
- Could not plan properly

His friends said he was "no longer Gage."

Why is this important?

It showed that:

- The frontal lobe controls personality, decision-making, and emotion regulation.

- Damage to the brain can change who we are, not just how we think.

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7. Language and the Brain (Aphasia)

Aphasia = language impairment from brain damage.

There are 3 important language areas:

1. Broca’s Area (Left Frontal Lobe)

Function

- Controls speaking
- Helps form words
- Coordinates speech muscles

Damage → Broca’s Aphasia

- Person understands speech
- BUT cannot speak fluently
- Speech becomes broken or slow
- Example: “Walk… dog… store…”

2. Wernicke’s Area (Left Temporal Lobe)

Function

- Understands language
- Gives meaning to words

Damage → Wernicke’s Aphasia

- Person can speak fluently
- BUT speech is meaningless
- Example sentence from a patient:
- “Mother is away her working her work to get her better…”

3. Angular Gyrus

Function

- Converts written text → auditory form
- Required for reading aloud

Damage

- Person can speak
- Person can understand speech
- BUT cannot read

8 How Reading Words Works (Flow of Processing)

1 → Visual Cortex Sees written words.

2 → Angular Gyrus

Converts written words into sound-like code.

3 → Wernicke's Area Interprets meaning.

4 → Broca's Area Plans speech.

5 → Motor Cortex

Moves the mouth to talk.

Example:

When reading the word "cat" aloud, your brain follows this exact pathway.

9. Thinking Styles (Left vs Right Brain Tendencies)

Not scientifically rigid, but helpful to understand preferences.

Intuitive / Creative Style

Thinks in images

Likes stories, patterns

Prefers hands-on learning

- Enjoys creative hobbies (art, music, writing)

Logical / Analytical Style

- Breaks problems into steps
- Prefers facts and sequences
- Likes instructions and guides
- Enjoys puzzles, math, strategy games

10. How to Enhance Brain Fitness (Based on Your Slides)

A. Sound Sleep

- Aim for 7.5 hours per night
- Sleep boosts memory, learning, mood, and focus
- Example: After studying, sleep helps store information more deeply.

B. Eat Well

Foods that support brain health:

- Nuts
- Blueberries

Whole grains

Avocados

Fish oil

- Broccoli
- Tomatoes
- Seeds

- Water

Nutrients important for the brain:

- Antioxidants (protect brain cells)
- Amino acids (help with neurotransmitters)
- Vitamin E (brain protection)

C. Write Things Down

- Helps clear your mind
- Reduces stress
- Improves memory
- Great habit: Write tomorrow's plan the night before

D. Slow Down

- Focus on one task at a time
- Spend time with people
- Build friendships
- Be mindful
- Practice empathy
    - Help helps your brain stay calm and organised

E. Breathe

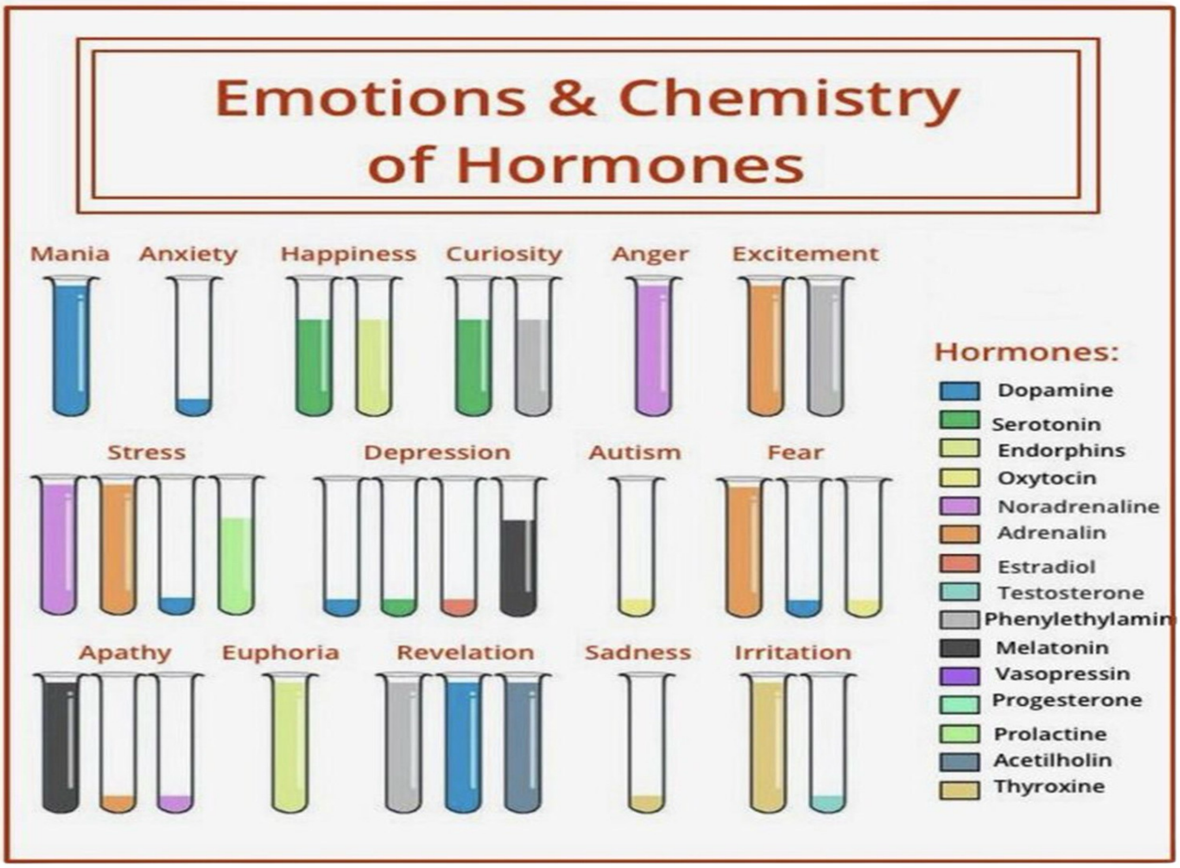
- Deep breathing reduces stress
- Improves oxygen flow to the brain
- Helps focus and emotional balance

Complete Summary (Quick Revision Section)

- Mind = what the brain does
- Brain controls everything: thoughts → emotions → senses → memory → body functions
- Neurons send electrical + chemical signals
- Brainstem = survival
- Limbic system = emotion, memory
- Cortex = thinking, planning, language
- Phineas Gage shows frontal lobe controls personality

- Broca's area = speaking
- Wernicke's area = understanding
- Angular gyrus = reading
- Brain fitness → sleep, nutrition, writing, slowing down, breathing

Neurotransmitter	Function	Examples of Malfunctions
Acetylcholine (ACh)	Enables muscle action, learning, and memory	With Alzheimer's disease, ACh-producing neurons deteriorate.
Dopamine	Influences movement, learning, attention, and emotion	Excess dopamine receptor activity is linked to schizophrenia. Starved of dopamine, the brain produces the tremors and decreased mobility of Parkinson's disease.
Serotonin	Affects mood, hunger, sleep, and arousal	Undersupply is linked to depression. Prozac and some other antidepressant drugs raise serotonin levels.
Norepinephrine	Helps control alertness and arousal	Undersupply can depress mood.
Endorphins	Lessen pain and boost mood	If flooded with artificial opiates, the brain may stop producing endorphins, causing intense discomfort.



Lecture 04: Emotion and Motivation

Part 1: Emotion

1. Definition of Emotion

Emotions are **complex psychological states** that involve three distinct components:

1. **Physiological arousal** – changes in the body (heart rate, hormones, sweating, etc.)
2. **Expressive behaviors** – outward signs of emotions (facial expressions, gestures, tone of voice)
3. **Conscious experience** – the internal, subjective feeling of the emotion (how you “feel” inside)

In simple terms:

Emotion = a biologically-based, learned, and personally relevant reaction involving both body and mind.

Example:

When you see a snake:

- Your heart rate increases (physiological response)
- You scream or jump back (expressive behavior)
- You feel fear (subjective experience)

2. Characteristics of Emotions

- **Biological basis:** Emotions are linked to brain structures (like the amygdala and hypothalamus).
- **Personal relevance:** Triggered by situations important to the individual.
- **Automatic and immediate:** Often occur instantly, before conscious thought.
- **Shaped by learning:** Experience and culture influence how we interpret and express emotions.
- **Adaptive value:** Emotions help humans survive by guiding behavior.

3. Components of Emotion

Component	Description	Example
Physiological	Body's response (autonomic nervous system activation)	Heart pounding, sweating when afraid
Behavioral	Expressive behavior or action	Smiling, frowning, crying
Cognitive	Thoughts, labels, and interpretation	"This situation is dangerous" → fear

4. Early Displays of Emotion (Infants)

- Even newborns display **basic emotional expressions**, showing that emotions are **innate**.
- Basic emotions appear within the first few months:
  - Joy → smiling
  - Anger → crying or screaming
  - Disgust → wrinkled nose
  - Fear → startled reflex
  - Sadness → whimpering, tears

Conclusion: Emotional responses are biologically programmed rather than learned through experience.

5. Emotions Across Cultures

- **Universality:** Research by *Paul Ekman* shows that facial expressions for six basic emotions are recognized across all human cultures.
- **Cultural variations:**
  - While emotions are universal, the **rules for expressing** them (called *display rules*) diPer.
  - Example:
    - In Japan, people hide anger in public to maintain harmony.
    - In Western cultures, people express emotions more openly.

6. Six Basic Emotions (Paul Ekman)

Psychologist **Paul Ekman** identified six emotions that are **innate and universal**:

Emotion	Description	Facial Expression Example

Happiness Feeling pleasure and satisfaction Smiling, relaxed muscles

Sadness Feeling loss or disappointment Drooping eyelids, tears

Fear Perceiving threat or danger Wide eyes, tense body

Disgust Feeling repulsion or aversion Wrinkled nose, pursed lips

Anger Response to threat or injustice Frowning, clenched jaw

Surprise Reaction to unexpected events Raised eyebrows, open mouth

These emotions are considered the foundation for more **complex emotions** like jealousy, pride, guilt, and shame.

---

7. Emotion vs. Mood

Aspect **Emotion** **Mood**

Duration Short-term (seconds to minutes) Long-term (hours to days)

Cause Specific trigger Often unclear or general **Aspect** **Emotion** **Mood**

Intensity Strong and focused Milder and diPuse **Example** Feeling angry after insult Feeling irritable all day **Example:**

You might feel **anger (emotion)** after losing a game, but that may turn into a **bad mood** lasting the whole evening.

---

8. Functions of Emotions

Emotions are **essential for human functioning** and serve three main purposes:

1. **Prepare us for action:**
   - Emotions trigger physiological responses that ready the body for response.
   - Example: Fear → "fight or flight" → increases heart rate and alertness.
2. **Shape future behavior:**
   - Emotional experiences help in **learning**.
   - Example: Fear of failure motivates you to study harder next time.
3. **Help in social interactions:**
   - Emotions communicate our feelings to others and promote bonding.
   - Example: A smile signals friendliness and approachability.

---

9. Biological Basis of Emotion

- **Amygdala:** Central in fear and aggression.

- **Hypothalamus:** Regulates physiological responses to emotion (e.g., sweating, heart rate).
- **Prefrontal cortex:** Controls emotion regulation and decision-making.
- **Autonomic Nervous System (ANS):** Produces changes in body state (sympathetic = arousal; parasympathetic = calm).

Part 2: Motivation

---

1. Definition of Motivation

Motivation is the **internal process** that initiates, guides, and maintains goal-directed behavior.

It is what **causes us to act** — whether to eat, study, exercise, or pursue dreams.

Key elements of motivation:

1. **Activation** – deciding to take action.
2. **Persistence** – continuing ePort despite obstacles.
3. **Intensity** – the amount of focus and energy applied.

---

2. Types of Motivation

Type	Description	Example
Intrinsic Motivation	Driven by internal satisfaction or interest.	Studying because you love learning.
Extrinsic Motivation	Driven by external rewards or avoiding punishment.	Studying to get good grades or avoid scolding.

Intrinsic motivation usually leads to **long-term satisfaction** and creativity.

---

3. Theories of Motivation

B. Instinct Theory (Evolutionary Perspective)

- Proposed by **William McDougall** and influenced by **Charles Darwin**.

- States that **behaviors are driven by instincts**, which are **inborn biological tendencies** that help survival.

- Humans have instincts like curiosity, parental care, and social belonging.

Limitation:

- Cannot explain learned or complex behaviors like career choice or studying for exams.

Example:

- A newborn's rooting reflex to find milk is instinctive.

C. Drive-Reduction Theory (Clark Hull)

- Based on the principle of **homeostasis** (balance).
- When a biological need (e.g., hunger, thirst, sleep) is unmet, it creates an **internal drive**.
- This drive **motivates** behavior to reduce tension and restore balance.

Process:

1. **Need:** Water
2. **Drive:** Thirst
3. **Behavior:** Drinking
4. **Homeostasis:** Restored balance

Limitation:

- Cannot explain behavior that increases tension (e.g., thrill-seeking, fasting).

D. Arousal Theory

- People seek to maintain an **optimal level of arousal** — neither too low nor too high.
- Too little arousal → boredom

- Too much arousal → stress
- Moderate arousal → peak performance (known as the *Yerkes-Dodson Law*) **Example:**
- A student performs best in exams when slightly nervous (optimal arousal).
- Too calm → lazy; too stressed → forgetful.

E. Maslow’s Hierarchy of Needs

- Proposed by **Abraham Maslow (1943)**.
- Suggests human needs are **arranged in a hierarchy** from basic to advanced.
- Lower needs must be met **before** higher ones become motivating.

Maslow’s 5 Levels of Needs

Level Need Type **Description Examples**

Basic biological Air, food, water, sleep, **Physiological**
Eating, sleeping needs sex, shelter

Safety Protection from harm, job Safe home, stable Security, stability
security, health income

Love &
Belongingness
Social Friendship, intimacy, Making friends, relationships family, group
belonging being loved

Esteem
Respect and Self-respect, recognition, Getting awards, achievement
confidence success

Self-
Realizing potential, Becoming an artist,
Actualization Personal growth creativity, fulfillment helping others

Key idea:

You cannot focus on higher needs (esteem/self-actualization) if lower needs (food/safety) are unmet.

Example:

A starving person will not care about fame until they find food.

---

4. **Applying Maslow's Hierarchy**

- Helps in **personal development**, **education**, and **workplace motivation**.
- Employers can motivate employees by ensuring:
  - Basic salary (physiological)
  - Job security (safety)
  - Team bonding (belongingness)
  - Recognition (esteem)
  - Growth opportunities (selfactualization)

---

5. **Comparison of Motivation Theories**

Theory	Key Idea	Motivation Source	Example
Instinct Theory	Behavior is innate instincts	Biological	Babies cry when hungry
Drive-Reduction	Behavior reduces internal tension	Biological drives	Drinking when thirsty
Arousal Theory	Seek optimal arousal level	Internal arousal	Doing sports for excitement
Maslow's Hierarchy	Needs must be met in order	Hierarchy of needs	Studying → Self-growth

---

---

6. Modern Understanding of Motivation

- **Cognitive factors:** Expectations, goals, and beliefs influence motivation.
- **Social factors:** Family, peers, and culture shape what we value and pursue.
- **Emotional link:** Positive emotions (joy, excitement) increase motivation, while negative emotions (fear, anxiety) may reduce it.

---

7. Connection Between Emotion and Motivation

- Emotions and motivation are **deeply linked**:

- Emotions **trigger** motivation (fear motivates escape; love motivates care).
- Motivation **influences** emotions (achieving goals brings joy).

- Both involve:
  - Biological activation (nervous system) o Cognitive interpretation o Behavioral response

---

8. Key Takeaways

- **Emotion** = Body + Mind reaction to meaningful events.
- **Mood** = Longer emotional state.
- **Emotions** help in adaptation, communication, and learning.
- **Motivation** is the internal energy that drives goal-oriented behavior.
- **Homeostasis** and **arousal levels** maintain internal balance.
- **Maslow's hierarchy** organizes needs from basic to self-fulfillment.
- Both emotion and motivation are **interconnected** processes that shape human behavior and personality.

LECTURE 05-SENSATION & PERCEPTION

1. INTRODUCTION: How We Experience the World

Your senses work **as a team** to help you navigate the environment.

Two main processes:

1. **Sensation** — receiving raw information.
2. **Perception** — interpreting the information to make sense of the world.

2. SENSATION

Definition:

Sensation is the **process by which sensory receptors and the nervous system detect and receive stimulus energy** from the environment.

How Sensation Works (Step-by-step):

1. A **physical stimulus** (light, sound, smell, touch, taste) hits a sensory organ.
2. Sensory receptors **absorb** this energy.
3. They **convert** the energy into **neural impulses** (electrical signals).
4. These impulses travel to the brain through neural pathways.

Key Idea:

Sensation = **Raw detection of stimulus** It does **not** yet have meaning.

3. PERCEPTION

Definition:

Perception is the **process of selecting, organizing, and interpreting sensory information** to form meaningful understanding of objects and events.

How Perception Works:

- The brain receives signals from sensation.
- It **organizes** (groups, separates, prioritizes).
- It **interprets** them based on experience, memory, and expectations.

Key Idea:

Perception = **Meaning-making**

Without perception, sensations are just meaningless signals.

---

4. HOW SENSATION & PERCEPTION WORK TOGETHER

Flow:

Stimulus → Sensory Organ → Sensation → Neural Messages → Brain → Perception → Understanding

Example:

- Sensation: Light hits the retina.
- Perception: Brain recognizes a face.

Diagram (text-version)

Stimulus → Receptors → Neural Signals → Brain → Interpretation → Perception

---

5. THRESHOLDS

Threshold = The dividing line between detectable and undetectable stimulation.

5.1 ABSOLUTE THRESHOLD

Definition:

The **minimum** amount of stimulation needed to detect a stimulus **50% of the time**.

Examples:

- Seeing a candle flame from 30 miles away on a dark night.
- Hearing a watch tick from 20 feet away.
- Smelling one drop of perfume in a three-room apartment.

5.2 DIFFERENCE THRESHOLD (Just Noticeable Diference – JND)

Definition:

The **smallest** change in stimulus intensity that a person can detect.

Important Rule — Weber's Law:

The stronger the original stimulus, the greater the change needed to notice a diference.

Examples:

- 5 → 10 lbs = noticeable
- 100 → 105 lbs = not noticeable
- TV volume 10 → 12 = noticeable
- TV volume 70 → 72 = barely noticeable

DiPerence threshold deals with **change**.

---

6. SIGNAL DETECTION THEORY (SDT)

(Very important for exam questions)

Purpose:

Explains how and why people detect faint signals diPerently depending on psychological and physical conditions.

Key Concepts:

- The brain always has **neural noise** (background random firing).
- A faint stimulus creates a **signal**.
- The brain must decide:

"Is this noise or is this a real signal?"

Detection depends on:

1. **Stimulus intensity**
2. **Person's state:**

o fatigue o
stress o
expectatio
n o motivation
o alertness

Outcomes (SDT matrix):

Actual Signal Response Example

hearing a sound that is actually
Present Hit
there failing to hear
someone call your
Present

M
i
s
s
n
a
m
e

thinking your phone vibrated when
Absent False Alarm it didn't

Absent Correct Rejection recognizing that a sound was not
real

Application Areas:

- Decision-making

- Medical diagnosis
- Security checking
- Radar operation

- Military alert systems

---

7. SENSORY ADAPTATION

Definition:

Reduced sensitivity to a constant or unchanging stimulus.

Why it happens:

The brain filters out irrelevant, constant information to focus on **new and important changes**.

Examples:

Becoming unaware of your clothes touching your skin.

- No longer smelling perfume after a few minutes.
- Chef not noticing kitchen heat.
- Worker ignoring construction noise.

---

8. VISION: HOW THE BRAIN MAKES SENSE OF SIGHT

Key Idea:

Vision is **not** just receiving light — it is the **brain's interpretation** of that light.

Why illusions occur:

Our brain evolved shortcuts to interpret the world quickly. These shortcuts work 99% of the time, but sometimes fail → illusion.

---

9. THE VISUAL PROCESS (STEP-BY-STEP)

Text Diagram:

Light → Cornea → Pupil → Lens → Retina → Optic Nerve

→ Thalamus → Visual Cortex (Occipital Lobe) **Important parts:**

- **Retina:** Converts light into neural signals.
- **Optic Nerve:** Sends signals to brain.

- **Visual Cortex:** Processes shapes, colors, movement, faces.

---

10. COLOUR VISION

Young–Helmholtz Trichromatic Theory

Main Idea:

The retina has **three types of cones**:

- **Red**
- **Green**
- **Blue**

All perceived colors = combinations of these three.

Examples:

- Red + Green = Yellow
- Green + Blue = Cyan • Red + Blue = Purple

---

11. COLOR BLINDNESS

Definition:

Reduced ability to see or distinguish between certain colors.

Most common type:

Red–Green color blindness.

Slide Example:

People with normal vision see **“74”** in the dot pattern.

People with color blindness may not.

---

12. VISUAL INFORMATION PROCESSING

12.1 Feature Detection (Hubel & Wiesel)

Definition:

Specialized neurons in the occipital lobe respond only to specific visual features:

- Edges
- Angles
- Lines
- Movements

These neurons help identify objects quickly.

12.2 Supercell Clusters (David Perrett)

Some brain cells respond to:

- Gaze direction
- Head angle
- Posture
- Body movement

These cells help:

- read social cues
- understand others' attention
- detect threats
- predict movement

---

13. PARALLEL PROCESSING

Definition:

The brain processes **multiple visual aspects at the same time**.

The four parts processed in parallel:

1. **Color**
2. **Motion**
3. **Form**
4. **Depth**

Example:

When watching a bird, you instantly know:

- its color
- shape
- how fast it’s moving
- how far away it is

Your brain does all these **simultaneously**.

---

14. SIMPLIFIED SUMMARY OF VISUAL PROCESSING

Memory Flowchart:

Retina detects light →

Feature detectors analyze edges and motion →

Supercells integrate movement + direction →

Parallel processing handles color, form, depth, motion →

Brain combines everything into meaningful vision

EXAM SURVIVAL SUMMARY (90-SECOND REVIEW)

- **Sensation** = detecting physical energy
- **Perception** = interpreting it
- **Absolute Threshold** = minimum detectable 50%
- **Diference Threshold (JND)** = smallest noticeable change
- **SDT** = detection depends on stimulus + person’s state
- **Sensory Adaptation** = lower sensitivity after constant stimulus
- **Vision Pathway** = Retina → Brain
- **Trichromatic Theory** = 3 cones (RGB)
- **Color Blindness** = dificulty distinguishing colors
- **Feature Detection** = specialized cells for edges, lines
- **Supercells** = gaze, posture, movement
- **Parallel Processing** = color, motion, form, depth simultaneously

---

LECTURE 06-Memory

Remembering, Forgetting, and How Memory Works

1. What Is Memory?

Definition

Memory is the mental system that **encodes**, **stores**, and **retrieves** information. It allows us to learn from experiences, recognize familiar people or objects, solve problems, and maintain our sense of identity.

Explanation

Memory is not like a video camera. It is an active process shaped by:

- **Bottom-up inputs** — sensory data (what we see/hear)
- **Top-down influences** — prior knowledge, expectations, culture, emotions

Mini Diagram

Experience → Encoding → Storage → Retrieval → Behavior/Decision

Example

You recognize your friend's voice (stored auditory memory) when they call you.

Case Example

A student remembers exam content better when they connect new concepts with old knowledge — showing topdown influence.

Memory Trick

Think **ESR** — Encoding, Storage, Retrieval (like "EdiSon's Record").

---

2. Bottom-Up vs Top-Down Influences on Memory

Bottom-Up Processing

Information flows from **sensation → perception → memory**.

Example:

You hear a loud bang. You remember the sound because it was surprising.

Top-Down Processing

Your memory is shaped by your **existing knowledge**, **culture**, and **expectations**.

Example:

If you believe cats are playful, you may remember a cat video as funnier than it was.

Cultural Insight

People from collectivist cultures tend to recall **social interactions** better, while individualistic cultures recall **personal achievements** more strongly.

Mini Diagram

BOTTOM-UP: Sensory → Meaning

TOP-DOWN: Meaning → Interpretation of Sensory

---

3. The Three Stages of Memory

3.1 Encoding

Definition

The process of transforming sensory input into a meaningful memory trace.

Types of Encoding

Type	Description	Example
Visual	Images	Remembering a logo
Auditory	Sounds	Remembering a melody
Semantic	Meaning-based	Remembering “democracy = people’s rule”

Psychological Insight

Deep, meaningful processing activates more brain areas → stronger memories.

3.2 Storage

Definition

Maintaining information over time.

Example

Your ability to remember multiplication tables years later.

3.3 Retrieval

Definition

Accessing stored information when needed.

Example

You recall your ATM PIN when withdrawing money.

Application

Self-testing strengthens retrieval pathways.

Mini Diagram

Encoding → (Storage) → Retrieval → Performance

---

4. Three-Stage Model of Memory

Sensory Memory → Short-Term Memory → Long-Term Memory

(attention) (rehearsal)

---

5. Sensory Memory

Definition

Brief storage of raw sensory information (visual, auditory, smell, taste, touch).

Types

Type	Duration	Example
Iconic (visual)	0.3–0.5 sec	Afterimage of a sparkler
Echoic (auditory)	2–4 sec	“What did you say? … oh yes.”
Haptic (touch)	1–2 sec	Feeling of a tap

Key Properties

- Very accurate but fades quickly
- Helps us perceive continuity (e.g., movies appear smooth)

Example Activity

“What letters did you just see?” — illustrates iconic memory.

---

6. Short-Term Memory (STM)

Definition

A temporary workspace where information is held for about **15–30 seconds**.

Capacity: “7 ± 2 chunks” (Miller, 1956)

Chunking

Meaningful grouping of information to improve capacity.

Example:

“17762024” → “1776 | 2024” (2 chunks instead of 8 digits)

Digit Span Test (from lecture)

Shows increasing chunk capacity from 4 to 11 digits.

How Prior Knowledge Helps

- Helps form meaningful chunks
- BUT may distort memory (top-down bias)

Mini Diagram

INFO → STM (limited) → rehearsal → LTM

---

7. **Long-Term Memory (LTM)**

Definition

Permanent, nearly limitless storage of knowledge, skills, and experiences.

7.1 Implicit (Non-declarative) Memory

Definition

Memory retrieved *without conscious awareness*.

Types & Examples

Type	**Example**
Procedural	Riding a bike
Priming	Seeing “doctor” makes you think of “nurse”
Conditioned responses	Fear of dogs after a bite

Mini Case

A car driver automatically uses brake+clutch timing — procedural memory.

7.2 Explicit (Declarative) Memory

Definition

Memory retrieved *with conscious eLort*.

Types

Type	**Description**	**Example**
Episodic	personal experiences	Your 10th birthday party
Semantic	facts, knowledge	Capital of France = Paris

Implicit vs Explicit — Comparison Table

Feature	**Implicit**	**Explicit**
Awareness	Unconscious	Conscious
Retrieval	Automatic	EPortful
Examples	Skills, priming	Facts, events
Brain Areas	Basal ganglia, cerebellum	Hippocampus

8. Efects That Influence Memory Encoding

8.1 Primacy and Recency Efect Explanation

In a list:

- **Primacy** → better recall of early items (LTM)
- **Recency** → better recall of later items (STM)

Example

You remember the first and last items in a shopping list better than the middle.

8.2 Levels of Processing (Craik & Lockhart, 1972)

Shallow Encoding

Focus on appearance (capital letters).

→ Poor memory.

Deep Encoding

Focus on meaning.

→ Strong memory.

Experiment Example

Participants recall more words when asked:

"Does this word relate to something alive?" than

"Is this word written in capital letters?"

Table

Encoding Type	Focus	Outcome
Shallow	Looks, sound	Weak memory
Deep	Meaning	Strong memory

8.3 Organization Matters

Semantically organized lists (e.g., "Career", "Home", "Food") lead to better recall than random lists.

Example:

Grouping "LAWYER, PHYSICIAN, TEACHER" under "PROFESSIONAL" improves memory

Mini Diagram

Random → weak memory

Organized → strong memory

8.4 Context-Dependent Memory

You recall better when in the same environment in which you learned.

Example:

Studying in your room → better recall during the exam if mental context matches.

---

9. **Forgetting**

9.1 Ebbinghaus Forgetting Curve

Memory loss is rapid at first, then plateaus.

Diagram (Text-based)

100% |■■■■■■■■■■■

80% |■■■■■

60% |■■■

40% |■■

20% |■

0% |________________

Time →

Example

You forget most of a lecture within days unless reviewed.

9.2 Interference Theory

Proactive Interference

Old information blocks new information.

Example:

You keep typing your old password when trying to memorize a new one.

Retroactive Interference

New information disrupts old information.

Example:

Learning a new phone number makes you forget your old phone number.

Table

Type	**Old vs New**	**Example**
Proactive	Old → interferes with new	Old password blocks new
Retroactive	New → interferes with old	New student names erase old

---

10. Flashbulb Memories

Definition

Extremely vivid, emotionally charged memories of dramatic events.

Features

- Caused by surprise + emotional arousal
- High confidence, but not always accurate
- Often distorted over time

Examples

- Remembering where you were during a major tragedy
- Recalling news of a national event

Cultural Insight

Communities often share “collective flashbulb memories,” strengthening group identity.

---

11. Improving Memory (Scientifically Proven Strategies)

1. Repeated, spaced study

→ Better than last-minute cramming.

2. Elaborative rehearsal

Connect with meaning, stories, personal examples.

3. **Self-reference efect** Link material to your life.

Example:

“Proactive interference is like when I forget my new Wi-Fi password.”

4. **Retrieval practice**

Testing yourself strengthens pathways.

5. **Reduce interference** Study in a quiet room.

6. **Sleep**

Consolidates memories into LTM.

7. **Exercise**

Boosts blood flow + neurogenesis.

8. **Organization**

Use diagrams, tables, categories.

9. **Brain-trainning apps**

Mildly helpful for attention and working memory.

10. **Healthy lifestyle**

Diet, socialising, staying mental active.

---

12. Final Summary Table — MEMORY (Cheat Sheet)

Topic	**Core Idea**	**Key Example**
Encoding	Transforming info into memory	Semantic > visual
Storage	Keeping info over time	STM 15–30 sec
Retrieval	Finding stored info	Self-testing
Sensory Memory	Brief sensory storage	Iconic = 0.5 sec
STM	7 ± 2 chunks	Phone numbers
LTM	Permanent store	Facts, skills
Implicit	Unconscious memory	Riding a bike
Explicit	Conscious recall	Birthdays
Primacy/Recency	Serial position ePects	Remember first & last

Topic	Core Idea	Key Example
Deep vs Shallow	Meaning > appearance	“Living?” > “CAPITAL?”
Interference	Old ↔ New conflict	Password confusion
Flashbulb	Emotional memories	Tragic events
Improving Memory	Spaced repetition, sleep, meaning	Study strategies

---

LECTURE 07: STRESS MANAGEMENT

“Understanding Stress, Trauma, and Self-Care”

(Full Master Note — Expanded, Clear, Human-Readable, Exam-Ready)

---

1. What is Stress?

Definition

Stress is the **psychological and physiological response** to any event or situation (a **stressor**) that disrupts the body’s equilibrium (balance).

Stress aPects:

- **Mind** (thoughts)
- **Emotions** (fear, anger, worry)
- **Body** (heart rate, breathing, hormones)
- **Behavior** (avoidance, aggression, poor performance)

Key Characteristics

- Stress can be *real* (a danger) or *imagined* (a worry).
- The brain activates the same response for both.

1.1 Why Zebras Don’t Get Ulcers (Sapolsky’s Idea)

Robert Sapolsky explains diPerences between animal and human stress:

Animals

- Stress is **acute** (short-term)
- Triggered by **physical threats**
- Ends once danger is gone

Example:

A zebra runs from a lion → either survives or becomes lunch → stress ends.

Humans

- Stress is **chronic** (long-term)
- Triggered by **thoughts, fears, expectations**
- We replay events repeatedly → stress stays ON

Example:

Thinking about an exam happening next month can cause stress TODAY.

Mini Diagram

Animal Stress → Physical → Short → Survival Based → Ends Quickly

Human Stress → Mental → Long → Future Focused → Stays Active

Real-Life Case Example

Rina missed one deadline. Instead of letting it go, she keeps thinking:

“What if I fail the course? What if my parents get angry?”

→ Stress continues for weeks

→ Sleep disturbance + headaches

This is **human anticipatory stress**.

---

2. Is Stress Always Bad?

(Weiss, Levine, Seligman)

Not all stress is harmful.

2.1 Types of Stress

Type	**Description**	**Example**
Eustress (Good Stress)	Motivates, energizes	Excitement before a match
Distress (Bad Stress)	Overwhelms, drains	Fear of failing
Acute Stress	Short and intense	Sudden braking while driving
Chronic Stress	Long-term, harmful	Years of financial pressure

---

3. Psychological Modifiers of the Stress Response

These factors determine how strongly a person reacts to stress.

3.1 Outlets for Frustration

Healthy release mechanisms:

- Exercise
- Art / music
- Talking to someone
- Journaling
- Walking

Example:

A student frustrated with workload goes for a 20-minute run → stress decreases.

3.2 Sense of Predictability

Predictable events → less stress

Unpredictability → increased stress

Example:

Getting exam routines early reduces anxiety.

3.3 Sense of Control

Feeling “I can handle this” lowers stress.

Example:

Making a study plan increases confidence.

3.4 Social Support

Friends and family reduce emotional load.

Case Example:

People recovering from surgery recover faster when they have supportive family visits.

Mini Formula

More Control + Predictability + Support + Outlets = Less Stress Impact

---

4. PTSD (Post-Traumatic Stress Disorder)

PTSD occurs after experiencing or witnessing trauma.

Definition

A mental health condition involving intense, disturbing thoughts and emotions related to a traumatic event.

Trauma Examples:

- Road accidents
- Violence / abuse
- Natural disasters
- Sudden death of loved ones
- War
- Severe medical procedures

---

5. Signs of PTSD (4 Symptom Clusters)

A. Intrusive Symptoms

- Flashbacks
- Nightmares
- Unwanted memories

- Emotional distress when reminded
- Physical reactions (sweating, shaking)

Example:

Hearing a loud bang and suddenly reliving an explosion scene.

B. Avoidance Symptoms

Avoiding:

- Thoughts
- Conversations
- Places
- People
- Activities connected to trauma

Example:

Avoiding the road where a fatal accident happened.

C. Negative Changes in Thinking & Mood

- Negative beliefs ("I'm worthless")
- Blaming self or others
- Persistent sadness or fear
- Loss of interest in hobbies
- Feeling emotionally numb
- DiPiculty remembering parts of trauma

Case Example:

A survivor of robbery stops trusting all strangers and avoids going out at night.

D. Arousal & Reactivity Symptoms

- Easily startled
- Hypervigilance (always alert)
- DiPiculty sleeping
- Irritability
- Angry outbursts
- Trouble concentrating
- Self-destructive behavior (speeding, drinking)

Children (under 6)

- Re-enacting trauma during play
- Frightening dreams

---

6. Identifying Inner Resources

Inner resources = personal strengths that help us cope.

Examples:

- Confidence
- Patience
- Emotional intelligence
- Social skills
- Optimism
- Problem-solving
- Faith or philosophical grounding

Example:

Positive self-talk (“I can handle this”) reduces anxiety during exams.

---

7. Myths About Self-Care

Myth	Reality	Example

Self-care is selfish	It prevents burnout	Taking a short break improves productivity
Self-care is temporary	Consistent habits make lasting changes	Daily 10-min meditation
Only for women	Needed for all humans	Men experience burnout too
Takes too much time	Even 5 minutes counts	Stretching, deep breathing
Anything soothing is selfcare	Some soothing acts are harmful	Drinking alcohol ≠ self-care
Same for everyone	Personalized	One person likes journaling; another prefers exercise

8. Self-Care Strategies for PTSD & Stress

8.1 Physical Self-Care

- Exercise
- Yoga
- Breathing exercises
- Healthy Sleep patterns

Example:

Watering plants every morning as a calming routine.

8.2 Emotional Self-Care

- Journaling
- Reflecting on feelings
- Forgiving self and others
- Healthy emotional expression

8.3 Mental Self-Care

- Reading something inspiring
- Practicing gratitude
- Learning new skills

8.4 Social Self-Care

- Talking to trusted friends
- Spending time with supportive people
- Joining groups or communities

8.5 Boundary Setting

Say "NO" to:

- Overcommitment
- Energy-draining situations
- Toxic people

Example:

Rejecting a late-night plan before an exam.

8.6 Professional Support

- Therapy
- Counseling
- Mental health workshops

Example:

Someone with recurring flashbacks attending cognitive behavioral therapy (CBT).

9. Memory Trick: "CARE PLAN" for Stress Management

C – Control (gain control)

A – Acceptance (accept what can't be changed)

R – Release (remove emotional pressure)

E – Express (talk / journal)

P – Predictability (prepare ahead)

L – Limit stressors

A – Ask for help

N – Nourish yourself (sleep, food, rest)

10. Final Summary Table

Topic	Key Idea	Example
Stress	Body's response to threats	Exam anxiety
Sapolsky	Animals = acute; humans = chronic	Zebra vs student
PTSD	Trauma-based disorder	Accident survivor
Intrusion	Flashbacks & memories	Nightmares after injury
Avoidance	Staying away from triggers	Avoiding hospitals
Negative Mood	Guilt, fear, numbness	Disconnection from friends
Hyperarousal	Sleep issues, irritability	Startling easily
Stress Modifiers	Control, predictability, outlets, support	Study routine reducing stress
Self-care	Healthy coping	Meditation, reading
Myths	Misunderstanding self-care	"Self-care is selfish" → false